

1) Database and DBMS are closely related but not same

* Database:- A database is an organised collection of data
It's where the actual information is stored.

* DBMS:- It is a software ~~between~~ that allows user and application to create, store, retrieve, update and manage the data of databases. It acts as interface between the user and database.

2) Advantages of DBMS:

- * A DBMS uses various powerful functions to store and retrieve data efficiently
- * Offers Data Integrity and Security.
- * Consumes lesser Space
- * Data Independence
- * Reduction of redundancy.

3) Database Instance:- It refers to the state of the data of datababase at a particulae moment of time.
→ It includes the actual stored data values

Database Schema:- It is a overall logical design (or) Structure of the database

→ It defines how data is organised.

4) → Representation of Strong Entity in E-R model

* By a single rectangle with entity name inside

Ex1- Student

→ Representation of Weak Entity in E-R model

* By a double rectangle with entity name inside

Ex1- Contact

5) Querying Relational data means retrieving specific information from relational database by asking queries with certain conditions. This is written in SQL [Structured Query language] & allows users to select, filter, join and manipulate the data.

Ex1- A Table of STUDENT [Name, Age, Course, Roll-No]

Sql: SELECT Name from STUDENT where Course = "CSE";

Justification1- This query access the relational data stored in STUDENT table and retrieve only the required information based on condition.

6) Selection (σ):- It selects rows that satisfies condition

Ex1- Relation: STUDENT (Name, Age, Roll-NO)

Name	Age	Roll-NO
Ajay	20	101
Ram	21	102

Relation Algebra : $\sigma_{Age > 20} (STUDENT)$

Result:-

Name	Age	Roll-NO
Ram	21	102

Projection (π):- Selects only column from relation

Ex-1 Relation: STUDENT (Name, Age, Roll-No)

Name	Age	Roll-No
Ajay	20	101
Ram	21	102

Relation Algebra: $\pi_{\text{Name}}(\text{STUDENT})$

Result:-

Name
Ajay
Ram

7, If two relation R and S:

* R has attributes (A, B)

* S has attribute (B)

Then $R \div S$ returns all values of A that are associated with every value of B in S.

Ex-1

Student	Course
Ravi	DB
Ravi	Java
Priya	DB

Relation R (StudentCourse):

Relation B (required Course):

Course
DB
Java

Relation Algebra:-

StudentCourse $\div$ required Course

Result:-

Student
Ravi

8> Domain Relational Calculus (DRC) is non-procedural query language in relational database. It focuses on what the data is required, not how to retrieve it. In DRC, it is expressed in logical formula with variables that represent domain (attribute value) rather than whole tuple.

9> Students A

StudentID	Name
101	Ravi
102	Priya
103	Arjun

Students B

StudentID	Name
102	Priya
103	Arjun
104	Meena

* UNION:
SELECT Name from StudentsA
UNION
SELECT Name from Students B

Name
Ravi
Priya
Arjun
Meena

* INTERSECT:
SELECT Name from StudentsA
INTERSECT
SELECT Name from Students B

Name
Priya
Arjun

* EXCEPT:
SELECT Name from StudentsA
EXCEPT
SELECT Name from Students B

Name
Ravi

10) Aggregate operators are operations in a database that are perform calculation on the set values and return a single summarized results. They are commonly used in relational algebra and SQL.

Ex:- COUNT, SUM, MAX, MIN, AVG